

Some definitions and terminology

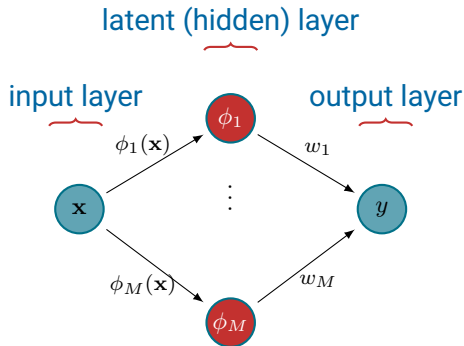
Narrow versus General AI:

- Narrow AI can only perform one specific task $\Leftarrow$ ML techniques live here
- General AI can perform a multitude of tasks and program itself $\Leftarrow$ just a dream (for now)

Adaptive basis functions

For now, only half of the model is trainable:

- Input to hidden **encoding** ($\phi_1 \cdots \phi_M$) fixed, hidden to output **decoding** ($\mathbf{w}$) trained
- What if we could also train the first half?



Neural Networks – Activation functions

For a given neuron, **forward propagation** happens in two steps:

- A **linear** combination of values from the previous layer:

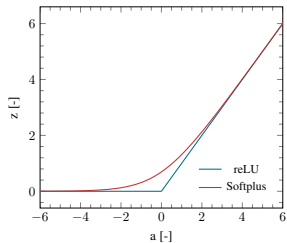
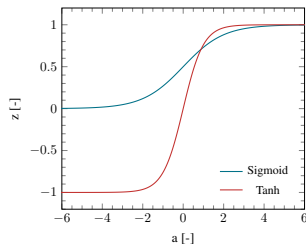
$$a_{lj} = \sum_i^D w_{ji}^{(l)} z_i^{(l-1)} + w_{j0}^{(l)}$$

- A **nonlinear** transformation with an **activation function**:

$$z_{lj} = h(a_{lj})$$

Choosing the activation function:

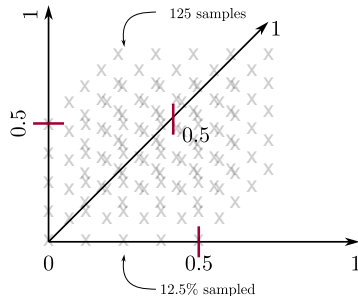
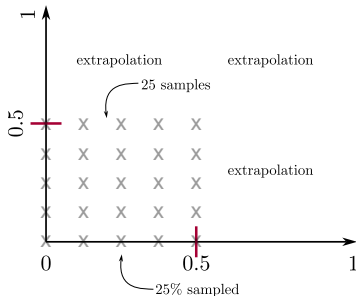
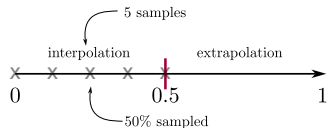
- Application dependent
- Can be seen as a **hyperparameter**



The Curse of Dimensionality

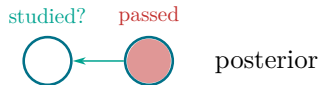
When building ML models, we rely on covering our feature space well enough. However...

- We need **exponentially more** samples to keep up as we go to higher dimensions



Bayes' Theorem – Example

Keeping our prior assumptions ignores important evidence. We use Bayes' Theorem instead:



$$p(S = 1|P = 1) = \frac{p(P = 1|S = 1)p(S = 1)}{p(P = 1)}$$